

| MedGuard — Defence Guide

Slide-by-slide script, source trace and question preparation

Ubaga Joshua Chukwuebuke · CSC/2022/009

Department of Computer Science, Chrisland University, Abeokuta

Supervised by Prof. Isaac Odun-Ayo · August 2026

HOW TO USE THIS DOCUMENT

There is one section per slide, twenty-nine in all. Each section has the same four parts:

What is on the slide — every element on the slide named and explained, including the ones you did not write yourself, like figure captions and the eyebrow label in the corner.

What to say — a spoken script. It is written to be said out loud, not read off. Times given are for a normal speaking pace; the whole deck runs to roughly twenty-two minutes if you keep to them.

Where the numbers come from — every figure on the slide traced to the file or paper it came from, so that if you are asked "where did that come from?" you have a one-sentence answer instead of a pause.

If they ask — the questions a panel is most likely to put to that slide, with answers. Answer in your own words; the point of the written version is that you have already thought it through.

Two conventions. Anything in *italics* is a stage direction, not something to say. Anything in **bold** inside the script is a number or phrase worth landing deliberately.

BEFORE YOU WALK IN

Five things to check, in order of how much damage they would do.

1. Know your participant groups cold. The deck and the final dissertation both say five physiotherapy students, six nursing students, four clinic staff. An earlier draft of the usability report labelled two of those groups differently, and your own August audit file still carries the old labels. If any old printout is circulating, destroy it. If asked who the fifteen were, the answer is: *five physiotherapy students, six nursing students, four clinic staff* — *no pharmacists and no prescribers, which is why it is listed as a limitation*.

2. The confusion matrix on slide 23. Cell counts are reconstructed from the per-tier precision, recall and support in `test_metrics.json`. They reconcile exactly to your reported 86.3% accuracy. The one cell not fully pinned by those metrics is True High → Predicted Low, shown as zero; its true value lies between 0 and 71. This is why the slide says "under 0.2% of High pairs fell to Low" rather than "none". If pressed on where the matrix came from, say it was reconstructed from the reported per-tier metrics and that it reconciles to the accuracy figure — do not claim the training script emitted it, because it does not.

3. The test count on slide 25. The slide says thirty-four automated tests. That figure is twelve backend test functions plus twenty-two mobile test *files*. Those twenty-two files contain one hundred and sixty-five individual test cases. So the slide understates rather than overstates, which is the safe direction — but know the breakdown before someone counts for you.

4. The database size. The slide says 79 MB. On disk the file is 75.1 MiB. Those are the same number: 75.1 mebibytes is 78.75 megabytes. If someone checks the file properties on a Linux machine and sees 75, that is the explanation.

5. Your one-sentence answer to "so what?" *No existing interaction checker is simultaneously free, offline and driven by a model rather than editorial consensus. MedGuard is all three, and it was tested with real users.*

THE ARGUMENT IN SIXTY SECONDS

If everything goes wrong and you have one minute, this is the spine:

Medication harm is common, expensive and largely preventable. The number of pairs a prescriber must check grows as the square of the medication count, and it grows fastest in exactly the settings with the fewest pharmacists. The existing tools are either paid, or free but online-only, and all of them assign severity by editorial judgement rather than from the pharmacology. I built a mobile application that ships its entire reference database inside itself, so it works with the network off; it grades uncatalogued pairs with a Random Forest trained on twenty-seven pharmacological features; and it recovers **91% of the interactions that matter most**. Fifteen users scored it **84.3** on the System Usability Scale. It is an academic prototype, not a clinically validated product, and the gap between those two things is the honest part of the work.

Slide 1 — Title

WHAT IS ON THE SLIDE

A split layout. The left two-thirds carries the text; the right third is a full-bleed panel with the application screenshot inside it.

Reading down the left: the eyebrow **FINAL YEAR PROJECT DEFENCE** in small caps; the full project title, *Implementation of a Machine Learning-Based Mobile Application for Drug Interaction Detection and Medication Safety Support*; a short accent rule; then a line of institutional text — Department of Computer Sciences, Chrisland University, Abeokuta, August 2026. Below a full-width hairline the slide splits into two credit blocks: PRESENTED BY with your name and matriculation number CSC/2022/009 on the left, SUPERVISED BY with Prof. Isaac Odun-Ayo and his role on the right.

WHAT TO SAY

Roughly twenty seconds. Do not read the title aloud — everyone can see it.

"Good morning. My name is Joshua Ubaga, matriculation number CSC/2022/009, and this is my final-year project, supervised by Prof. Odun-Ayo. The project is called MedGuard. It is a mobile application that checks a patient's medicines against each other and grades how dangerous each combination is, using a machine-learning model — and it does all of that with the network switched off."

WHERE THE NUMBERS COME FROM

Nothing on this slide requires sourcing. The one thing to check before the day is that the title here matches the title on the dissertation's certification page exactly; an earlier version of the document dropped the opening words "Implementation of a", and that was corrected in the August revision.

IF THEY ASK

"**Why is it called MedGuard?**" — It guards the medication list rather than the individual drug. The unit the application reasons about is the whole regimen, not one prescription at a time, and the name was chosen to carry that.

"**Is this deployed?**" — No. It is a working prototype: it builds, installs and runs on an Android device, and the backend is deployable, but it has not been released and it has not been clinically validated. Slide eight sets out exactly what that means.

Slide 2 — Presentation outline

WHAT IS ON THE SLIDE

The eyebrow reads **PRESENTATION OUTLINE**; the slide heading is "What this presentation covers". Beneath it, ten numbered cards in two columns of five. Column one, items 01 to 05: opening context; background of the study; statement of the problem; aim and objectives; significance of the study. Column two, items 06 to 10: review of related work; existing systems and research gaps; system design and methodology; implementation and results; conclusion and recommendations.

The structure deliberately mirrors the five chapters of the dissertation. Items 01–05 are Chapter One, 06–07 are Chapter Two, 08 is Chapter Three, 09 is Chapter Four, and 10 is Chapter Five.

WHAT TO SAY

Fifteen seconds. Point at the two columns; do not read all ten items.

"I will move through this in the order of the dissertation. The left column is Chapter One — why the problem matters and what I set out to do. The right column takes us through the literature, the design, the results, and what I would recommend next. I will spend most of the time on the right-hand column, and particularly on the results."

WHERE THE NUMBERS COME FROM

Not applicable. The mapping to chapters is worth stating out loud because it tells the panel you know your own document's structure.

IF THEY ASK

"How long will you take?" — Around twenty minutes, with the results section taking the largest share.

Slide 3 — Opening context

WHAT IS ON THE SLIDE

Three statistic cards across the full width, each with a large figure and a supporting sentence, and a source line beneath them reading "World Health Organisation, 2021 and 2023".

Card one: **1 in 30** — patients experience medication-related harm, and more than a quarter of those cases are severe or life-threatening. Card two: **\$42bn** — lost each year to unsafe medication practice, close to one per cent of global health spending. Card three: **50%** — of all preventable patient harm is caused by medicines, and most of it is avoidable.

The slide heading states the first figure in words: "One patient in every thirty is harmed by the medicines meant to help them."

WHAT TO SAY

Forty seconds. This is the emotional opening; slow down.

"Three numbers to frame the problem, all from the World Health Organisation.

One patient in every thirty is harmed by their own medication. More than a quarter of those cases are severe or life-threatening.

That harm costs about **forty-two billion dollars a year** — close to one per cent of everything the world spends on health.

And **half of all preventable patient harm** comes from medicines. The word that matters there is preventable. This is not an unavoidable cost of treatment. It is a checking problem, and checking is something software is good at."

WHERE THE NUMBERS COME FROM

All three are World Health Organisation figures, from the *Global patient safety action plan 2021–2030* (WHO, 2021) and the 2023 medication-safety materials. The reference list on slide 28 carries the 2021 citation.

The "more than a quarter are severe or life-threatening" clause qualifies the 1-in-30 figure and is part of the same WHO source, not a separate estimate.

IF THEY ASK

"Are these global figures relevant to Nigeria?" — They are global, and that is why the very next slide narrows to Nigerian data specifically. The WHO figures establish that the problem is large; the Nigerian figures on slide four establish that it is worse where support is thinner.

"One per cent of global health spending — is that a meaningful proportion?" — It is the WHO's own framing of the forty-two billion figure. Its purpose here is scale, not precision: it says medication harm is a line item large enough to matter at national-budget level.

"Preventable by what?" — The WHO attributes most of it to system factors rather than individual error, and interaction checking is one of the named system controls. That is the specific slice this project addresses; it does not address the others, such as dispensing or administration error.

Slide 4 — Background of the study

WHAT IS ON THE SLIDE

This is the densest slide in the deck. It has three regions.

Left panel — "Pairs a prescriber must check." A small bar chart with four rows: 2 drugs → **1** pair; 5 drugs → **10**; 8 drugs → **28**; 10 drugs → **45**. Beneath it, a note: polypharmacy runs at 37% generally and 45% among older adults, and DrugBank 6.0 documents 1.4 million interactions across 11,900 drugs.

Top-right — two cards. 0.53 pharmacists per 10,000 people. **74.7%** of health spending paid out of pocket.

Middle-right — two cards. 50.5% of 430 Benin City admissions carried an interaction on day one; **67.4%** carried one by the time they were discharged.

Bottom-right — a wide card, "Web tools fail at the counter": connectivity in community pharmacies and rural clinics is unreliable, yet 90.7% of Nigerians have mobile access. The phone is already in the room.

WHAT TO SAY

Ninety seconds. This slide carries the whole justification; take your time and use the three regions in order.

"Now to why this is hard, and why it is hardest here.

On the left is the arithmetic of the problem. The number of pairs a prescriber must check is n times n -minus-one over two. Two drugs is one check. Five drugs is ten. Ten drugs is **forty-five separate checks** — and this is not an unusual patient. Polypharmacy, five or more medicines at once, runs at thirty-seven per cent generally and **forty-five per cent among older adults**. Meanwhile DrugBank documents one-point-four million known interactions. No clinician holds that in their head.

Top right is the supply side. Nigeria has about **half a pharmacist per ten thousand people**, and almost **three-quarters of health spending is paid out of pocket**. So the workload rises and the professional support to absorb it is not there.

Then the middle two cards show what that produces in practice. A study of four hundred and thirty admissions at the teaching hospital in Benin City found that **half of patients arrived already carrying a drug interaction** — and by the time they were discharged, **two-thirds did**. The hospital stay made it worse.

And the last card is why this had to be a mobile application rather than a website. Connectivity at the counter is unreliable. But **ninety per cent of Nigerians have mobile access**. The device is already in the room. The problem is that the software on it assumes a network."

WHERE THE NUMBERS COME FROM

The pair counts are the combination formula $n(n-1)/2$, computed directly: $2 \rightarrow 1$, $5 \rightarrow 10$, $8 \rightarrow 28$, $10 \rightarrow 45$. There is no citation because it is arithmetic; say so if asked.

Polypharmacy 37% and 45% — Delara et al. (2024), an umbrella review of polypharmacy prevalence covering 1997 to 2022, in *Archives of Gerontology and Geriatrics*. On slide 28.

1.4 million interactions across 11,900 drugs — Knox et al. (2024), DrugBank 6.0, *Nucleic Acids Research*. The precise figure in your dissertation is 1,413,413 interactions. An earlier draft carried two conflicting versions of this (14,000 drugs / 2.6 million interactions); it was reconciled to one figure in the August revision.

0.53 pharmacists per 10,000 — Ekpenyong et al. (2018), a census-based analysis of Nigeria's pharmacy workforce, *Journal of Pharmaceutical Policy and Practice*. Your dissertation gives the range as 0.53 to 0.66; the slide uses the lower bound. The same paper found 21,892 registered pharmacists with only 59% in active practice.

74.7% out-of-pocket — national health-expenditure data cited in Chapter One.

50.5% and 67.4% — Oriaifo et al. (2024), *Clinical and Translational Science*, a study of 430 admissions to the medical wards at the University of Benin Teaching Hospital. On slide 28.

90.7% mobile access — Nigerian telecommunications penetration data cited in Chapter One.

IF THEY ASK

"Does every one of those forty-five pairs actually need checking?" — No, and that is the honest answer. Most pairs are clinically irrelevant, which is precisely the argument for automation: the machine can dismiss the forty-one that do not matter so the clinician's attention goes to the four that do. What a human cannot do is know in advance which four.

"The Benin City study is one hospital. Is that generalisable?" — It is not, and I do not claim it is. It is used as an existence proof that the problem is present and measurable in a Nigerian tertiary setting, not as a national prevalence estimate. The national-level claims on this slide come from workforce and expenditure data, which are census-based.

"Why 0.53 when your dissertation says 0.53 to 0.66?" — The lower bound is the more conservative figure for the point being made, which is scarcity. Either number is far below the WHO density threshold.

"Mobile access is not the same as smartphone ownership." — Correct, and that is a fair limitation. The 90.7% figure is mobile access broadly. MedGuard requires an Android smartphone, so the addressable population is smaller than that number implies. What the figure establishes is that the mobile channel is the realistic one, not that every one of those users could run this application today.

Slide 5 — Statement of the problem

WHAT IS ON THE SLIDE

Three cards across the width, then a full-width statement bar underneath.

Cost — the most accurate platforms are subscription products priced for well-funded hospitals, out of reach for most public-sector facilities and individual practitioners in Nigeria.

Connectivity — free checkers remove the cost barrier but need a live connection for everything they do. At the point of care in a rural clinic, that is exactly when they stop working.

Translation — machine learning classifies interactions well in published work, but Wang et al. (2024) found most models stay on benchmarks and never reach an application anyone can open.

The bar beneath states the gap: no available tool combines offline-first operation, machine-learning-based risk prediction and multiple safety checks in one application built for a low-resource setting.

WHAT TO SAY

Sixty seconds. The three cards are three separate failures; make that structure audible.

"So why has this not been solved? Three separate barriers, and they do not overlap.

Cost. The most accurate platforms — Lexicomp is the reference standard — are subscription products priced for well-funded hospitals. They are simply out of reach for most Nigerian practice.

Connectivity. The free tools remove that barrier, but every one of them needs a live connection for everything it does. Which means they fail at exactly the moment you need them: at the counter, in a rural clinic, with no signal.

Translation. And in the research literature, machine learning does classify interactions well — but Wang and colleagues, reviewing the field in 2024, found that most of these models never leave the benchmark. They are never built into anything a person can open.

Put those together and the gap is specific: **no available tool is offline-first, model-driven and multi-check at the same time**, built for a setting like this one. That is what this study addresses."

WHERE THE NUMBERS COME FROM

There are no statistics on this slide; it is an argument. Its one citation, **Wang et al. (2024)**, *Machine learning for drug–drug interaction prediction: a comprehensive review*, is on slide 28. The cost and connectivity claims are substantiated on slides 10 and 11, where the four tools are compared directly.

IF THEY ASK

"Isn't 'no available tool does all three' a very strong claim?" — It is bounded by the four systems I reviewed in Chapter Two, which are the ones a Nigerian practitioner would realistically reach for: Lexicomp, Medscape, Drugs.com and DDInter. Slide 11 shows the comparison. I do not claim to have surveyed every product in existence, and I would state it as: none of the four most commonly used checkers does all three.

"Why is offline the binding constraint rather than cost?" — Because cost has an obvious solution and offline does not. There are already free checkers, so cost alone is solved. What is not solved is that free and offline have

never been combined, and the technical reason is that the free tools are all thin web clients over a server-side database.

Slide 6 — Aim and objectives

WHAT IS ON THE SLIDE

A tall card on the left holds the **AIM OF THE STUDY**: to build a mobile application that detects drug–drug interactions and grades their risk for patients and healthcare providers.

The right-hand side is headed "Specific objectives", annotated "Five, in order", and lists five numbered cards:

- 01 — Build the Flutter application on a bundled offline database.
- 02 — Train and compare classifiers that predict a risk tier.
- 03 — Integrate food, duplicate therapy, allergy and dose checks.
- 04 — Serve the model as an API and distil it for on-device use.
- 05 — Evaluate usability with users in Abeokuta.

WHAT TO SAY

Forty-five seconds.

"The aim is one sentence: build a mobile application that detects drug–drug interactions and grades their risk.

Underneath it are five objectives, and they are deliberately in dependency order. **One**, build the application on a database that ships inside it. **Two**, train and compare classifiers to predict a risk tier — note *compare*, not just train; the comparison is part of the objective. **Three**, add the other safety checks: food, duplicate therapy, allergy, dose. **Four**, serve that model two ways — as an API for when there is a network, and distilled onto the device for when there is not. **Five**, put it in front of real users and measure.

Every one of these is answered by a specific results slide later, and I will point at them as we go."

WHERE THE NUMBERS COME FROM

These are the objectives as stated in Chapter One of the dissertation. The phrase "Five, in order" is a design annotation, not a claim.

IF THEY ASK

"**Which objective was hardest?**" — Objective four. Getting a Random Forest to run on a phone is not possible directly, so it had to be distilled into a small neural network and converted to TensorFlow Lite. The result is under eight kilobytes and agrees with the forest 95.6% of the time, and slide 25 reports that.

"**Did you meet all five?**" — Yes, all five are evidenced in Chapter Four. Objective five is met but narrowly: I evaluated usability with fifteen users, none of whom was a pharmacist or prescriber, which is a limitation I state on slide eight rather than hide.

"**Why compare classifiers at all rather than pick the best-known one?**" — Because the choice had to be justified against deployment constraints, not just accuracy. Slide 18 shows the comparison and slide 9 explains why the strongest published approach, graph neural networks, was excluded before training began.

Slide 7 — Significance of the study

WHAT IS ON THE SLIDE

Three tall cards, one per beneficiary group.

Patients on several medicines — immediate, evidence-based warnings about risky combinations, particularly for older adults who carry the highest interaction risk.

Practitioners without support tools — decision support that needs no hospital infrastructure and no record-system integration, so judgement is spent where it matters.

The research field — a demonstration that a severity model can travel from training all the way to a working application built on free, open data.

WHAT TO SAY

Thirty-five seconds.

"Who this is for, in three groups.

Patients on several medicines — and especially older adults, who carry the highest interaction risk — get an immediate, evidence-based warning rather than finding out after the fact.

Practitioners without support tools get decision support that requires no hospital infrastructure and no integration with any records system. It runs on a phone that is already in their pocket.

And for **the research field**, the contribution is the pipeline itself: a demonstration that a severity model can travel all the way from training to a working application, on nothing but freely licensed data and software. Wang and colleagues identified that as the thing the field mostly does not do."

WHERE THE NUMBERS COME FROM

No statistics. The "older adults carry the highest interaction risk" claim is supported by the 45% older-adult polypharmacy figure on slide 4 (Delara et al., 2024).

IF THEY ASK

"Is a patient the right user for this? Could it cause self-medication?" — This is the sharpest question on this slide and it deserves a direct answer. The design mitigates it three ways: the application never recommends stopping or changing a medicine, the High-tier wording is "seek professional advice" rather than any instruction, and a disclaimer is shown at first launch. But the risk is real and it is one of the reasons I recommend pharmacist review before any wider release.

"What does 'free, open data' mean when DrugBank requires an academic licence?" — Fair correction, and my dissertation had that contradiction in an earlier draft. DrugBank is free for academic use under licence, not open in the public-domain sense. RxNorm, OpenFDA and DDInter are openly accessible. The honest phrasing is "freely licensed for academic use".

Slide 8 — Limitations of the study

WHAT IS ON THE SLIDE

Four cards across the width, then a full-width clarifying note.

Not clinically validated — no institutional trial and no regulatory review by NAFDAC or the FDA. It supports professional judgement; it does not replace it.

No pharmacist or prescriber tested it — the cohort was students and clinic staff in one city, so it says nothing about the clinician the system is partly designed for.

Never load-tested — latency was measured one request at a time. Nothing is known about the prediction service under concurrent use.

The shipped database lags the model — 13.9% of its records would move from Moderate to Major under the current classifier. DrugBank suspended academic downloads before it could be rebuilt.

The note beneath: the model results are unaffected, because the classifier was trained on the corrected labels. It is the database that fell behind, not the model.

WHAT TO SAY

Sixty seconds. Deliver this without apology — volunteering limitations before you are asked is the strongest position you can be in.

"Before the results, what this study does not claim. There are four.

It is not clinically validated. There has been no institutional trial and no review by NAFDAC or the FDA. It supports professional judgement; it does not replace it.

No pharmacist or prescriber tested it. My fifteen participants were students and clinic staff in one city. So the usability result says nothing about the clinician this system is partly designed for.

It was never load-tested. I measured latency one request at a time. I do not know how the prediction service behaves under concurrent use, and I am not going to pretend otherwise.

And **the shipped database lags the model.** About **fourteen per cent** of its records would move from Moderate to Major under the current classifier. I could not rebuild it, because DrugBank suspended academic downloads before I could.

One clarification on that last point, because it matters: **the model results are unaffected.** The classifier was trained on the corrected labels. It is the database that fell behind, not the model."

WHERE THE NUMBERS COME FROM

13.9% — the proportion of records in the shipped `medguard.db` whose severity label would change from Moderate to Major under the current labelling rules, reported in Chapter Four.

The DrugBank suspension is an external event; it is the reason the rebuild is recommendation three on slide 27 rather than something completed.

The cohort description — five physiotherapy students, six nursing students, four clinic staff — is Table 4.9 of the final dissertation. No pharmacists, which is what makes this limitation true.

IF THEY ASK

"If the database is wrong, are your results wrong?" — No, and this is the key distinction. The model was trained and evaluated on corrected labels, so every number on slides 22 to 24 stands. The stale records are in the copy bundled with the application, which affects what a user of the prototype would see for those specific pairs, not what the classifier learned. It is a deployment defect, not a validity defect.

"Fourteen per cent sounds like a lot." — It is, and it is why rebuilding the database is a recommendation rather than a footnote. It is also directional: those records move Moderate → Major, meaning the shipped database *under*-warns relative to the model. If it erred the other way I would be more worried.

"Why did you not use a different data source once DrugBank closed?" — The labels come from DDInter, which is open, but the *features* are DrugBank-derived — enzymes, half-life, ATC classes. Substituting a different pharmacological source would have meant rebuilding the feature pipeline and retraining from scratch, which was not possible in the time remaining. That is the honest reason.

"You say it is not a replacement for a professional. Is that enforced anywhere in the software, or is it just a disclaimer?" — Partly enforced. The application never issues an instruction to stop or change a medicine; the strongest thing it says is "seek professional advice". But it is fundamentally a disclaimer plus a wording discipline, not a technical control, and I would not claim more than that.

Slide 9 — Review of related works

WHAT IS ON THE SLIDE

A four-column comparison table — Approach, Strength, Cost, Fit for this study — with four rows.

Graph neural networks — strongest published benchmark accuracy; needs a knowledge graph and GPU-class compute; cannot run on a phone.

Deep multimodal models — combine structure, targets, enzymes and pathways; opaque, and need a second model to explain each score; too heavy, hard to audit.

Tree ensembles — native feature importances, strong on tabular features; slightly behind deep models on raw accuracy; deployable and interpretable.

Shared weakness — all three degrade on newly approved drugs; severity datasets are dominated by moderate cases; measured here in Chapter Four.

Source line: Wang et al. (2024) · Xia et al. (2025) · Zhang et al. (2023) · Pham et al. (2024).

WHAT TO SAY

Seventy seconds. The fourth row is the important one; make sure you reach it with time to spare.

"Three families of approach dominate the literature, and I want to be clear that I did not choose the most accurate one.

Graph neural networks produce the strongest published benchmark accuracy. They also need a curated knowledge graph and GPU-class compute. They cannot run on a phone, so they were excluded before training began — not because they are worse, but because they fail the deployment constraint.

Deep multimodal models combine molecular structure, targets, enzymes and pathways. They are opaque: you need a second model just to explain each score. For a clinical tool that has to be auditable, that is a serious cost.

Tree ensembles are slightly behind deep models on raw accuracy, but they give feature importances natively — the model explains itself — and they deploy. That is what I chose, and slide 18 shows the comparison I ran.

But the row that matters most is the last one. **All three families share the same two weaknesses.** They all degrade on newly approved drugs, and severity datasets everywhere are dominated by moderate cases. I did not fix either of those. What I did do is **measure both of them**, on slides 23 and 24, which the literature mostly does not."

WHERE THE NUMBERS COME FROM

There are no statistics on this slide — it is a synthesis of four reviews and papers, all in the reference list:

Wang et al. (2024) — the comprehensive review of ML for DDI prediction; the source for the benchmark-to-application gap and for the family comparison.

Xia et al. (2025) and **Zhang et al. (2023)** — the graph and deep-model performance claims.

Pham et al. (2024) — cited for the class-imbalance characterisation.

An earlier draft of the dissertation cited Deng et al. (2020) and Yu et al. (2021) here for a claim already supported by Xia and Zhang; those were removed in the August revision as inherited placeholders.

IF THEY ASK

"You excluded the best-performing approach. Isn't that a weakness?" — It is a constraint, deliberately applied, and the dissertation states it as such. The research question is not "what is the most accurate DDI classifier" — that question is answered in the literature. It is "what can be deployed offline on a phone and still be useful". A graph neural network that cannot run on the target device scores zero on that question regardless of its benchmark accuracy.

"How much accuracy did you give up?" — I cannot quantify it directly, because I did not train a GNN to compare. The published gap in Xia et al. and Zhang et al. is a few percentage points on their own benchmarks, which are not my dataset or my label scheme, so any number I gave would be a false comparison. What I can say is that within the three algorithms I did train on identical features and splits, the spread was 0.486 to 0.562 macro-F1.

"Measured here in Chapter Four' — where exactly?" — Cold-start degradation is slide 24; class imbalance shows up as the weak Low tier on slides 22 and 23. Both are reported rather than smoothed over.

"Why not use a pre-trained chemical language model?" — Same constraint. The embedding step alone exceeds the device budget, and the resulting scores are not interpretable without a secondary explanation model, which is the objection in row two.

Slide 10 — Review of existing systems

WHAT IS ON THE SLIDE

Four cards, each with a screenshot of the tool above and three text blocks below: name, access model, and an assessment.

Lexicomp · Subscription — first for accuracy and completeness. Priced beyond most Nigerian practice, and its severity ratings are editorial rather than derived.

Medscape · Free, online only — no registration, three severity levels. Scored 202 of 400 for accuracy against Lexicomp's lead, with abbreviated descriptions.

Drugs.com · Free, ad-supported — a credible free alternative that has matched subscription tools on some interaction sets. Thin clinical depth, no offline mode.

DDInter 2.0 · Free, open access — pharmacist-reviewed ratings across about 240,000 interactions. Web only. MedGuard uses its ratings as an expert override layer.

WHAT TO SAY

Sixty seconds.

"These are the four tools a practitioner would actually reach for.

Lexicomp is the reference standard — first for accuracy and completeness in the comparative literature. Two problems: it is priced beyond most Nigerian practice, and its severity ratings are *editorial*. A committee assigns them. That is not wrong, but it is not derived from the pharmacology either.

Medscape is free and needs no registration, with three severity levels. In a head-to-head accuracy study it scored **two hundred and two out of four hundred** against Lexicomp's lead, with abbreviated descriptions.

Drugs.com is a credible free alternative that has matched subscription tools on some interaction sets, but the clinical depth is thin and there is no offline mode.

DDInter 2.0 is the interesting one. It is open access, with pharmacist-reviewed ratings across roughly **two hundred and forty thousand** interactions. It is web-only, so it fails the same connectivity test as the others — but its ratings are good enough that MedGuard uses them as an **expert override layer**: DDInter can raise an alert the model missed, but it can never suppress one."

WHERE THE NUMBERS COME FROM

202 of 400 — a comparative accuracy scoring study of interaction checkers, cited in Chapter Two, in which Lexicomp led the field.

~240,000 interactions — DDInter's published coverage. Chapter Two cites Xiong et al. (2022) for the original release and Tian et al. (2025) for version 2.0. Note that the August revision corrected the tool's name here: an earlier draft called it "DDI Web Checker 2.0", which is not its name.

The screenshots are the four unnumbered figures in Chapter Two. They had captions but no figure numbers in an earlier draft and did not appear in the List of Figures; that was corrected.

IF THEY ASK

"What does 'editorial rather than derived' mean, and why is it a criticism?" — In Lexicomp, a clinical committee decides that a given pair is a level-2 or level-3 interaction. That is expert judgement, and expert judgement is valuable. But it is not reproducible from inputs, it cannot be audited by anyone outside the committee, and it cannot generalise to a pair nobody has reviewed. A derived severity — one computed from enzyme overlap, half-life, therapeutic index and so on — can do all three. That is the third research gap on slide 11.

"If DDInter is free and pharmacist-reviewed, why not just ship DDInter?" — Because it is web-only and covers about 240,000 interactions. MedGuard ships 851,868 interaction records offline and can generate a prediction for a pair that appears in neither. I do use DDInter — as the gold-standard labels for training and as the override layer at runtime.

"An override that can raise but never suppress — why that asymmetry?" — Because the two errors are not symmetric in a safety tool. Suppressing a true alert can harm a patient; raising a false one costs attention. So the expert layer is allowed to make the system more cautious and never less. The same logic drives the threshold tuning on slide 23.

Slide 11 — Comparison of existing systems

WHAT IS ON THE SLIDE

A five-column table — System, Access model, Strength, Limitation, Offline — with a row for each of the four tools. The Offline column reads **No** four times, which is the entire point of the slide.

Lexicomp · Subscription · detailed monographs and mechanisms · costly, editorial severity assignment · No.

Medscape · Free · no registration, multi-drug checks · limited clinical depth · No.

Drugs.com · Free, ad-supported · easy to use, broad public access · consumer-level detail only · No.

DDInter 2.0 · Open access · pharmacist-reviewed severity ratings · web only, no mobile application · No.

Below the table, three gap cards:

GAP 1 — No affordable checker works without a network.

GAP 2 — Interaction models stay in the laboratory.

GAP 3 — Severity rests on editorial consensus, not features.

WHAT TO SAY

Fifty seconds. Land the "No" column with a pause.

"The same four tools, side by side, on five dimensions. Access model, strength, limitation — and the last column.

[pause] **Offline. No. No. No. No.**

That column is the finding. These four systems differ in price, in depth and in audience, and on the one dimension that matters at the point of care in a rural clinic, they are identical.

From that comparison, three gaps. **One:** no affordable checker works without a network. **Two:** interaction models stay in the laboratory — Wang and colleagues' finding, and it is a gap between research and practice, not a gap in research. **Three:** severity rests on editorial consensus rather than on features you can inspect.

Those three gaps are what the rest of this presentation answers, one at a time."

WHERE THE NUMBERS COME FROM

The table is a synthesis of the four systems reviewed in Chapter Two. Every cell traces back to the tool's own documentation or to the comparative studies cited on slide 10.

The three gaps are sections 2.8.1 to 2.8.3 of the dissertation, restated in section 2.9. An earlier draft of Chapter Five mistakenly said Chapter Two identified six gaps; that was corrected to three.

IF THEY ASK

"Some of these have mobile apps. Doesn't Medscape have one?" — Medscape has a mobile application, but the interaction checker within it queries a server. The distinction I am drawing is not "has an app" but "works with the network off", which requires the reference data to be resident on the device. None of the four does that.

"Three gaps from four tools is a small evidence base." — The four were selected as the ones in actual use in the target setting rather than as a systematic sample, and I would not defend them as exhaustive. But gaps one and three are structural rather than sampling artefacts: any thin web client fails gap one by construction, and any human-curated severity scale fails gap three by definition.

"Is gap two really a gap, or just a commercialisation lag?" — It is partly that. But Wang et al.'s point, and mine, is that the lag is systematic rather than incidental — the modelling choices that maximise benchmark scores are the same ones that make deployment impossible. That is why this project treated deployability as a design constraint from the start rather than as a later engineering problem.

Slide 12 — Research gaps

WHAT IS ON THE SLIDE

A large figure fills the left two-thirds: **Figure 1, Identified Research Gaps and Proposed System Features**. It maps each gap to the feature that answers it.

On the right, three cards restating that mapping:

No genuine offline mode → answered by shipping the whole database inside the app, so the network becomes an enhancement rather than a requirement.

Models never reach a user → answered by carrying the model all the way: trained, served through an API, distilled onto the device, and tested with real users.

Severity rests on editorial consensus → answered by deriving severity from pharmacology, with expert ratings kept as an override that can raise an alert but never suppress one.

WHAT TO SAY

Fifty seconds.

"This slide is the hinge of the presentation: three gaps on the left, and what each one demands of the system on the right.

Gap one, no genuine offline mode. Answered by shipping the entire database inside the application. Not caching, not sync — the whole thing, resident. Which turns the network from a requirement into an enhancement.

Gap two, models never reach a user. Answered by carrying the model the whole distance: trained, served through an API, distilled down onto the device, and then actually tested with real people. Any one of those steps missing and it is still a benchmark result.

Gap three, severity rests on editorial consensus. Answered by deriving severity from the pharmacology — enzymes, half-life, therapeutic index — and keeping the expert ratings as an override that can raise an alert but never suppress one.

Everything after this slide is the execution of those three answers."

WHERE THE NUMBERS COME FROM

No statistics. Figure 1 is your own diagram, from Chapter Two of the dissertation. Note that it is captioned Figure 1 on the slide but carries its chapter number in the document — check which numbering the panel is reading from if a figure is referred to by number.

IF THEY ASK

"**Shipping the whole database — how big does that get, and does it scale?**" — 79 MB for 4,955 drugs and 851,868 interaction records, which is a normal app-store download. It does not scale indefinitely: the compression work described on slide 17 was what made it viable, and a full DrugBank at 1.4 million interactions would need more aggressive measures. That is a real ceiling on the approach and I would state it as one.

"**'Raise but never suppress' — could that flood the user with alerts?**" — It could, and alert fatigue is a documented failure mode in clinical decision support; my Chapter Two cites work finding physicians override

roughly ninety per cent of interaction alerts. My mitigations are that Low-tier results raise no alert at all, and that the report orders findings by severity rather than listing them flat. But I have not measured alert fatigue, because that requires longitudinal use, and I would not claim to have solved it.

Slide 13 — Research methodology

WHAT IS ON THE SLIDE

Two bands. The upper band is the five-phase Design Science Research process, laid out left to right as five connected boxes:

PHASE 1 Problem and requirements → PHASE 2 Data engineering pipeline → PHASE 3 Model training and selection → PHASE 4 Application and API build → PHASE 5 Testing and user evaluation.

The lower band is the three evaluation strands:

Experimental — the classifier evaluated on a held-out test set and on a harder drug-disjoint split, against a majority-class baseline.

Functional — automated tests over every safety check, including a group that runs with the network disabled.

Usability — fifteen participants, five task scenarios each, and the ten-item System Usability Scale.

WHAT TO SAY

Sixty seconds.

"The methodology is Design Science Research, from Hevner and colleagues. The premise is that in this kind of work you *build* the artefact and then evaluate it rigorously — the construction is part of the method, not preliminary to it.

Five phases across the top. Problem and requirements; the data engineering pipeline; model training and selection; the application and API build; then testing and user evaluation.

And three strands of evaluation underneath, because one kind of evidence would not be enough.

Experimental — the classifier on a held-out test set, and then on a harder split where the drugs themselves are held out, both against a majority-class baseline.

Functional — automated tests over every safety check, including a group that deliberately runs with the network disabled, because that is the study's central architectural claim and it needs testing rather than asserting.

Usability — fifteen participants, five task scenarios each, and the standard ten-item System Usability Scale."

WHERE THE NUMBERS COME FROM

Design Science Research — Hevner, March, Park and Ram (2004), *MIS Quarterly*. On slide 28.

Fifteen participants, five scenarios — $15 \times 5 = 75$ task instances, which is the denominator in the "74 of 75" figure on slide 26.

The System Usability Scale — Brooke (1996) for the instrument, Bangor, Kortum and Miller (2008) for the interpretation bands. Both on slide 28.

The majority-class baseline is 76.1% (from `test_metrics.json`, `baseline_majority_accuracy: 0.7609`) — the accuracy you get by predicting "Moderate" for every pair.

IF THEY ASK

"**Why Design Science rather than an experimental or survey design?**" — Because the output of the study is an artefact. DSR is the framework built for exactly that situation: it treats the construction of the artefact as a

legitimate research contribution provided it is rigorously evaluated and the design decisions are justified against a stated problem. A pure experimental design would have had nothing to experiment on.

"Fifteen participants — how did you arrive at that number?" — Chapter Three originally specified five to ten, on the basis of Nielsen and Landauer's work on defect discovery. I recruited fifteen, which exceeds that. For SUS specifically, fifteen is above the commonly cited minimum of twelve for a stable mean, though it is still small and the confidence interval is wide.

"What is a drug-disjoint split, and why did you do it?" — In a normal random split, the same drug can appear in both training and test pairs, so the model may have learned that drug's risk profile. In the drug-disjoint split, a drug appears in one side or the other but never both, so every test pair involves at least one medicine the model has never seen. It simulates a newly approved drug. Slide 24 reports both, and the gap between them is the honest cost of the feature design.

"Is a majority-class baseline a fair comparison?" — It is the weakest defensible baseline and I use it as a floor, not as a competitor. It matters here because the classes are so imbalanced that predicting "Moderate" every time already gets 76.1%. Without that baseline stated, 86.3% accuracy would look far more impressive than it is. I also ran McNemar's test to confirm the improvement is not chance.

Slide 14 — Conceptual framework

WHAT IS ON THE SLIDE

Figure 2, Conceptual framework of the MedGuard system, occupying the left two-thirds. It shows four layers with data flowing downward.

On the right, an explanatory card: DrugBank, RxNorm, OpenFDA and DDInter are parsed into one SQLite database. That database feeds two consumers — the model, which learns severity from pharmacological features, and the application, which bundles the same file for offline checking.

Below it, a card headed **EVERYTHING ENDS IN ONE OF THREE TIERS**, with three chips: Low, Moderate, High.

WHAT TO SAY

Fifty-five seconds.

"This is the whole system on one diagram, and the shape to notice is that it narrows.

At the top, four public data sources. **DrugBank** for pharmacology — enzymes, half-lives, ATC classes. **RxNorm** to normalise names, so that a brand name and a generic resolve to the same medicine. **OpenFDA** for adverse-event signal. And **DDInter** for expert-reviewed severity labels.

All four are parsed into **one SQLite database**. And that single database then feeds two different consumers: the **model**, which learns severity from pharmacological features, and the **application**, which bundles the very same file for offline checking. One source of truth, two uses — which is what keeps the model and the app from drifting apart.

And at the bottom, everything — every path, every check, catalogued or predicted — converges on **one of three tiers**: Low, Moderate, High. The user never sees a probability or a score. They see one word."

WHERE THE NUMBERS COME FROM

The four sources are as stated. **DDInter provides the gold-standard labels**; DrugBank provides the features. This division matters and is worth stating: it means the labels and the features come from different organisations, which reduces circularity.

The three tiers. The underlying source data has four severity grades — Minor, Moderate, Major, Contraindicated. The model's `tier_map` in `model_meta.json` collapses these to three by mapping both Major and Contraindicated to High. Catalogued pairs retain their original four-grade source label internally; the three tiers are the user-facing presentation.

IF THEY ASK

"**Why three tiers rather than four?**" — Two reasons. Clinically, Major and Contraindicated lead to the same user action: seek professional advice. Statistically, Contraindicated is so rare that keeping it separate would have produced a class the model could not learn. Collapsing them keeps the action space honest and the classes trainable. Note that an earlier draft of section 2.1.3 said the app uses four grades; that was a documentation error, corrected in August.

"Doesn't collapsing lose information?" — At the user-facing layer, yes, deliberately. The underlying grade is retained in the database for catalogued pairs, so nothing is destroyed — it is a presentation decision, not a data decision.

"OpenFDA is adverse-event reports. Isn't that noisy?" — Very. It is spontaneous reporting with well-known reporting bias, so it is used as a supporting signal rather than as a label source. The labels come from DDInter, which is pharmacist-reviewed.

"Why SQLite rather than a server database?" — Because it is the only mainstream database that can be shipped as a single file inside a mobile application and queried locally with no server process. That choice is what makes the offline claim possible; every other architectural decision follows from it.

Slide 15 — System architecture

WHAT IS ON THE SLIDE

Figure 3, High-level offline-first architecture of the MedGuard system, on the left.

On the right, two cards. The first: presentation, checking logic, the 79 MB reference database and an 8 KB model all sit on the device. Only an uncatalogued pair reaches the network, and only as two drug identifiers.

The second, headed **FALLBACK ORDER**: local database → cloud model → on-device model. The feature degrades; it never fails.

The slide heading states the claim: "The phone holds everything it needs. The cloud is optional."

WHAT TO SAY

Sixty seconds. This is the architectural centre of the project.

"The architecture, and the sentence at the top is the claim: the phone holds everything it needs, and the cloud is optional.

Everything sits on the device. The interface, the checking logic, the **seventy-nine megabyte** reference database, and an **eight kilobyte** model. Nothing leaves the phone unless the pair is not in the database — and when it does, what leaves is **two drug identifiers**. Not a patient record, not a medication list, not a name. Two identifiers.

The fallback order is the part I would draw your attention to. First we try the **local database**. If the pair is not there and there is a network, we ask the **cloud model**. If there is no network, we use the **on-device model**.

So the behaviour degrades — the on-device model is a distillation and it is slightly less accurate than the forest, and slide 25 quantifies that at ninety-five-point-six per cent agreement. But at no point does the feature fail. There is no state in which the user asks whether two drugs interact and gets nothing back."

WHERE THE NUMBERS COME FROM

79 MB — the shipped `mobile/assets/db/medguard.db`. On disk it is 75.1 MiB, which is 78.75 MB in decimal megabytes; both are the same file. An earlier draft claimed the database was slimmed "from roughly 295 MB to about 152 MB"; those figures could not be reproduced from the build script and were replaced with the measured 79 MB in the August revision.

8 KB — `severity_mlp.tflite`, precisely 7,984 bytes per `parity_report.json`.

The fallback order is implemented in the mobile client and is verified by the offline test group referenced on slide 25.

IF THEY ASK

"What exactly leaves the device?" — Two drug identifiers, in a single POST to the prediction endpoint. No user identifier, no medication list, no allergy data, nothing about who is asking. The user database is a separate, encrypted store that never leaves the phone.

"Why keep a cloud model at all if the device has one?" — Because the device model is a distillation and is measurably less faithful — 95.6% agreement with the forest. When a network is available, using the full Random Forest is strictly better. The on-device model is the floor, not the default.

"How does the bundled database get updated?" — Through a version-and-delta endpoint on the backend, which the API tests cover. It is implemented but it has not been exercised at scale, and it is not something I would claim as validated.

"Eight kilobytes seems impossibly small for a neural network." — It is a small multilayer perceptron over twenty-seven input features, trained on the forest's output probabilities rather than on the raw labels, then quantised through TensorFlow Lite. It has no embedding tables and no convolutional layers, which is where the size normally goes. The fidelity cost is measured, not assumed: 95.6% agreement, and 88.0% High-tier recall against the forest's 91.0%.

Slide 16 — System flowchart

WHAT IS ON THE SLIDE

Figure 4, System flowchart of the MedGuard safety-check process, on the left, showing the decision path for a single check.

Three explanatory cards on the right:

A catalogued pair returns its curated severity grade straight from the bundled database.

An uncatalogued pair goes to the model: cloud when there is a network, on-device when there is not.

Food, duplicate-therapy and allergy checks then run, and the verdict is the highest tier present.

WHAT TO SAY

Fifty seconds.

"One safety check, start to finish.

A pair arrives. First question: **is it in the bundled database?** If yes, we return its curated severity grade directly — no model involved, no network. That is the majority of checks, and it is why the application feels instant.

If the pair is **not** catalogued, it goes to the model. Cloud when there is a network, on-device when there is not — the fallback from the previous slide.

Then, whichever path produced the interaction verdict, **three more checks run**: food conflicts, duplicate therapy, and allergies.

And the final verdict is the **highest tier present** across all of them. If the interaction is Moderate but the patient has a recorded allergy to one of the drugs, the report comes back High. The most serious finding governs, and nothing gets averaged away."

WHERE THE NUMBERS COME FROM

No statistics. The "highest tier present" rule is implemented in the mobile safety-report logic and is covered by the automated tests on slide 25.

IF THEY ASK

"Why does a catalogued pair not go through the model too?" — Because a pharmacist-reviewed rating from DDInter is better evidence than a prediction. The model exists to cover pairs where no such rating exists. Using the model to override curated data would be replacing stronger evidence with weaker.

"Then what is the model actually doing most of the time?" — Covering the tail. The database holds 851,868 interaction records over 4,955 drugs, but the combinatorial space of pairs is much larger than that, so uncatalogued pairs are common in practice — particularly for less common medicines, which is exactly where a clinician is least confident and most needs support.

"Taking the maximum rather than combining the risks — is that not crude?" — It is deliberately crude, and I would defend it on two grounds. There is no validated model for combining a drug interaction with a food conflict and an allergy into a joint risk, so any combination rule would be invented. And taking the maximum is the conservative choice: it can never produce a verdict lower than the most serious individual finding.

Slide 17 — Entity relationship diagram

WHAT IS ON THE SLIDE

Figure 5, Entity–relationship diagram of the MedGuard reference database, on the left — one `drugs` table at the centre with six related tables around it.

On the right, an explanatory card: every spoke holds a foreign key back to `drugs`, normalised to third normal form. Repeated descriptions are stored as integer codes and rebuilt through a SQL view, which is what keeps the file small enough to ship.

Beneath it, four statistic tiles: **4,955** drugs · **851,868** interaction records · **252,046** synonyms for search · **79 MB** shipped inside the app.

WHAT TO SAY

Fifty-five seconds.

"The database schema. One `drugs` table at the centre, six spokes around it, every spoke carrying a foreign key back to the centre, normalised to third normal form.

The four numbers underneath are what actually ships. **Four thousand nine hundred and fifty-five drugs. Eight hundred and fifty-one thousand interaction records. Two hundred and fifty-two thousand synonyms**, which is what makes search tolerant of brand names and misspellings. And **seventy-nine megabytes** in total.

That last number is the engineering result. Eight hundred and fifty thousand interaction records with full text descriptions do not fit in seventy-nine megabytes. The reason they do here is that the descriptions are enormously repetitive — the same mechanism sentence recurs across thousands of pairs. So repeated text is stored **once, as an integer code**, and reassembled at query time through a SQL view. The application still sees full sentences; the file on disk stores integers."

WHERE THE NUMBERS COME FROM

All four verified directly against the shipped `mobile/assets/db/medguard.db`:

`drugs` — 4,955 rows. `interactions_data` — 851,868 rows. `drug_synonyms` — 252,046 rows. File size — 75.1 MiB, which is 78.75 MB, reported as 79 MB.

The database also holds 65,784 drug-category rows, 20,822 drug-enzyme rows and 2,503 food interactions. The lookup tables that do the compression are `tmpl_lookup` (718 templates), `mech_lookup` (15 mechanisms) and `effect_lookup` (13 effects). Full-text search indexes exist over both `drugs` and `drug_synonyms`.

Note: an earlier draft of the dissertation stated the schema had ten relations while the table listed nine, and placed the user tables in the reference schema. The corrected position is six reference relations plus three in the separate encrypted user store.

IF THEY ASK

"Third normal form — can you justify that?" — Every non-key attribute depends on the primary key and nothing else. Drug names live only in `drugs`; synonyms live in `drug_synonyms` with a foreign key; enzyme associations are a separate relation because a drug has many enzymes and an enzyme has many drugs. There are no repeating groups and no transitive dependencies.

"Storing text as integer codes and rebuilding through a view — isn't that denormalisation?" — It is the opposite. Repeating an identical description string across ten thousand rows is the redundancy that normalisation exists to remove. Extracting it into a lookup table with a foreign key is textbook normalisation; the only unusual part is that the target is a description rather than an entity.

"Where do allergies and user medications live?" — In a separate encrypted database on the device, `medguard_user.db`, never in the shipped reference file. That separation is why the reference database can be distributed with the application and the user's data cannot leave the phone.

"Only 4,955 drugs when DrugBank has 11,900?" — Correct. The shipped set is the subset with enough pharmacological detail to build the twenty-seven-feature vector — the drugs with enzyme, half-life and ATC data present. A drug with no such data could not be featurised, so including it would produce a record the model cannot score.

Slide 18 — The machine learning pipeline

WHAT IS ON THE SLIDE

A left card holding a horizontal bar chart of three classifiers, each with its validation macro-averaged F1:

Random Forest — 0.562 (bar in full teal, the winner)

Gradient Boosting — 0.548

Logistic Regression — 0.486

Caption beneath: validation macro-averaged F1 — same features, same splits.

Two cards on the right. The dark one: twenty-seven features per pair — shared enzymes, inhibition profiles, ATC-class overlap, half-life, narrow-therapeutic-index flags and each drug's recorded risk history. The white one: the forest was chosen for the highest macro-F1, a firmer hold on the minority High tier, and feature importances a clinician can interrogate without a second model to explain the first.

WHAT TO SAY

Seventy seconds.

"Three classifiers, trained on identical features and identical splits, so the comparison is clean.

Random Forest, point five six two macro-averaged F1. **Gradient Boosting, point five four eight. Logistic Regression, point four eight six.**

The forest wins, but by a narrow margin over gradient boosting — fourteen thousandths — and I want to be honest that this margin alone would not justify the choice. Three things together did.

It has the highest macro-F1. It holds the **minority High tier** more firmly, which is the tier that matters clinically. And it gives **feature importances natively** — the model explains itself, without needing a second model to interpret the first.

Note that I am reporting **macro-averaged F1**, not accuracy. That is deliberate: macro-averaging weights all three tiers equally, so a model cannot score well by simply predicting the dominant class. With classes this imbalanced, accuracy would flatter every one of these three.

On the right: **twenty-seven features** per drug pair. Shared metabolic enzymes, inhibition profiles, ATC-class overlap, half-life ratios, narrow-therapeutic-index flags, and each drug's recorded risk history — which turns out to matter enormously, and that is the next slide."

WHERE THE NUMBERS COME FROM

All three figures from `ml_pipeline/models/test_metrics.json`, field `model_comparison_val_macro_f1`: RandomForest 0.5615, HistGradientBoosting 0.5484, LogisticRegression 0.4858. The slide rounds to three decimal places.

These are **validation** scores, not test scores — the model selection was made on the validation split and the test set was only touched afterwards. That is the correct discipline and worth saying if you get the chance.

Training configuration, from `train_model.py` and `model_meta.json`: 1,019,738 training rows, `class_weight="balanced"`, random state fixed for reproducibility. The tie-breaking rule in the code keeps the Random Forest if it is within 0.005 of the best model rather than switching on a trivial difference.

IF THEY ASK

"Fourteen thousandths is not a real difference. Did you test whether it is significant?" — Not between the two models directly, and that is a fair criticism. What I did test with McNemar's is the selected model against the majority-class baseline, where chi-squared is 6,823.8, comfortably significant. Between the forest and the boosted model I would not claim a statistically established difference, which is exactly why the selection rests on the interpretability and minority-class arguments as well.

"Why macro-F1 rather than accuracy or weighted F1?" — Because the Moderate class is 76% of the data. Weighted F1 and accuracy both let a model score highly while ignoring the High tier entirely, which is the one clinically consequential class. Macro-averaging treats a failure on the 433 Low pairs as seriously as a failure on the 166,000 Moderate ones.

"What is class_weight='balanced' doing, and did you consider resampling instead?" — It reweights each class inversely to its frequency during training, so the rare classes are not drowned out. I did not use SMOTE or other synthetic oversampling: with over a million training rows, generating synthetic pharmacological feature vectors risks manufacturing drug pairs that are chemically meaningless. Reweighting achieves the same objective without inventing data.

"Twenty-seven features seems arbitrary. How were they chosen?" — They are the features derivable for every drug pair from the four data sources without imputation beyond explicit missingness flags. They fall into groups: enzyme and mechanism features, ATC-class features, pharmacokinetic features, risk-history features, and a set of clinical flags. The count is the consequence of the data available, not a target.

"Did you tune hyperparameters?" — The forest uses balanced class weights and a fixed random state, and the decision thresholds were tuned on the validation set — High at 0.40 and Low at 0.75. I did not run an exhaustive grid search over tree depth and count, which is a limitation. A more thorough search might have closed or widened the gap to gradient boosting.

Slide 19 — Implementation: home and search screens

WHAT IS ON THE SLIDE

Two phone screenshots on the right, captioned **Home dashboard** and **Drug search**. Two text cards on the left.

The first: the dashboard opens on the current status of everything the user is taking. Drugs are added through autocomplete over 252,046 synonyms, so a brand name or a misspelling still resolves to one normalised medicine.

The second: adding or removing a medicine re-screens the whole list immediately. There is no separate check button to forget, and the lookup runs against the bundled database with no network call.

Heading: "The regimen is the unit of safety, not the single drug."

WHAT TO SAY

Fifty seconds.

"Now the application itself, and the design principle in the heading: **the regimen is the unit of safety, not the single drug**.

The dashboard opens on the current status of everything the user is taking — not a blank search box. The question the app answers on launch is 'am I safe right now', not 'what would you like to look up'.

Drugs go in through autocomplete over **two hundred and fifty-two thousand synonyms**, so a brand name, a generic name or a common misspelling all resolve to the same normalised medicine. That is the RxNorm normalisation doing its work, and it matters because a user who types a brand name and gets nothing back will not type again.

And the behaviour I would draw attention to: **adding or removing a medicine re-screens the entire list immediately**. There is no separate check button. A check button is a thing a user forgets to press, and a safety tool that depends on remembering to press something is not a safety tool. Every one of those lookups runs against the bundled database, with no network call."

WHERE THE NUMBERS COME FROM

252,046 synonyms — the `drug_synonyms` table in the shipped database, verified directly.

"No network call" — verified by the offline test group described on slide 25, which runs the checking paths with connectivity disabled.

IF THEY ASK

"What happens if a drug genuinely isn't in your database?" — Search returns no match and the drug cannot be added. That is a real limitation of a 4,955-drug catalogue, and the user is told the medicine is not recognised rather than being shown a silent empty result.

"Re-screening the whole list on every change — is that expensive?" — It is $n(n-1)/2$ lookups against a local indexed SQLite database. For a ten-drug regimen that is forty-five queries, which is well inside the local performance target. It would not scale to hundreds of medicines, but no patient takes hundreds.

"Did users understand the dashboard?" — Task completion on the dashboard scenarios was clean. The one task that produced difficulty across participants was finding the allergy profile, which is on slide 26 and is the basis of recommendation one.

Slide 20 — Implementation: safety report and regimen risk

WHAT IS ON THE SLIDE

Two screenshots on the right, captioned **Safety report** and **Regimen risk panel**.

On the left, a text card explaining that interactions, food conflicts, duplicate therapy and allergies resolve into a single verdict equal to the highest tier present, and that the report then says what to do about it, ordered by how much it matters.

Below that, three small grouped tiles showing the tier scheme:

- 1 • **Low** — no alert raised.
- 2 • **Moderate** — monitor or separate doses.
- 3 • **High** — seek professional advice.

And a closing card: severity is carried by the fill of the dial, the colour and the wording together, so it still reads for a user who cannot tell red from amber.

WHAT TO SAY

Fifty-five seconds.

"The safety report. Four separate checks — interactions, food conflicts, duplicate therapy, allergies — resolve into **one verdict**, equal to the highest tier present. And then the report says what to *do*, ordered by how much it matters.

The three tiers each carry an action rather than a label. **Low raises no alert at all** — that is deliberate, and it is why the model's weakness on the Low tier costs so little in practice. **Moderate** says monitor or separate the doses. **High** says seek professional advice — note that it does not say stop taking it, because that is not a decision this application is entitled to make.

And one accessibility point I want to make explicitly. Severity is carried **three ways at once**: by the fill of the dial, by the colour, and by the wording. So it still reads for a user with red–green colour deficiency. Colour alone would have been a defect in a safety tool, and roughly one man in twelve is affected."

WHERE THE NUMBERS COME FROM

The three-tier action mapping is the user-facing design specified in Chapter Three. The redundant-encoding decision (fill, colour and wording together) is a design choice justified against accessibility guidance in the same chapter.

IF THEY ASK

"**Why does Low raise no alert?**" — Because an alert with no action attached trains the user to ignore alerts. Low means no clinically meaningful concern, so surfacing it would add noise and erode attention for the Moderate and High cases. It also happens to be the reason the model's poor Low-tier performance is tolerable: a tier that produces no user-facing output is a tier whose errors do not reach the user.

"**Isn't 'seek professional advice' unhelpfully vague?**" — It is deliberately bounded. The application is not clinically validated and has had no pharmacist review, so any more specific instruction would exceed what the evidence supports. Making the wording more directive is something I would only do after the pharmacist review in recommendation four.

"Did you test the colour-blind case?" — Not with a colour-blind participant, no. The redundant encoding is a design mitigation applied from guidance, not a validated accessibility result, and I would not overstate it.

Slide 21 — Implementation: dose scheduler and reminder editor

WHAT IS ON THE SLIDE

Two screenshots, captioned **Dose calendar** and **Reminder editor**.

Two text cards. The first: dose scheduling, reminders, refill tracking and a QR emergency card — every one of these runs against the bundled database, with the network switched off.

The second: these features are what give the app a reason to be opened between checks, which is what keeps the medication list current enough for the checks to mean anything.

WHAT TO SAY

Forty-five seconds.

"Safety does not end at the check, so the application carries dose scheduling, reminders, refill tracking and a QR emergency card. All of them run locally, with the network off.

The second card is the argument for why these exist at all, and I think it is the more interesting point. These are **retention features**, and retention is a safety mechanism here rather than a product metric.

An interaction checker is only as good as the medication list it is checking. If the user opens the app once, enters four drugs and never returns, the list goes stale and every subsequent check is wrong. The reminders give a reason to open the app daily — which is what keeps the list current enough for the checks to mean anything.

The QR emergency card is the same reasoning at the other end: it makes the medication list useful to a clinician in an emergency, when the patient may not be able to recite it."

WHERE THE NUMBERS COME FROM

No statistics. The offline claim for these features is covered by the same automated offline test group as the checking paths.

IF THEY ASK

"**Are these in scope for a drug interaction project?**" — They are, on the retention argument: the checking accuracy depends on list currency, and list currency depends on the user returning. But I would accept that they are secondary contributions and they are not what the evaluation focuses on.

"**Did you measure whether reminders actually improve adherence?**" — No. That needs a longitudinal study with real patients over weeks, which was outside both the scope and the ethical approval for this work. The adherence logic is implemented and unit-tested; its clinical effect is unmeasured.

"**Is the QR card a privacy risk?**" — It encodes a medication list, which is health information, and anyone who can see the screen can read it. It is user-initiated rather than always-on, but it is a genuine trade-off between emergency utility and privacy, and I have not formally assessed it.

Slide 22 — Results: model performance

WHAT IS ON THE SLIDE

Four headline statistic cards across the top:

91.0% of High-risk interactions correctly flagged.

86.3% accuracy, against a 76.1% baseline.

0.943 macro AUC-ROC across the three tiers.

218,518 held-out drug pairs in the test set.

Two interpretation cards beneath:

It over-flags, on purpose — precision on the High tier is 0.658. In a decision-support tool, a false alarm is the safer error.

The Low tier is weak, and rare — just 433 of 218,518 pairs. A Low prediction raises no alert, so the practical cost is small.

WHAT TO SAY

Eighty seconds. This is the headline results slide; do not rush it, and do not skip the two cards at the bottom.

"The headline results, on **two hundred and eighteen thousand held-out drug pairs**.

The number I would lead with is the first one: **ninety-one per cent of High-risk interactions correctly flagged**. That is recall on the tier that matters clinically — of the interactions that could seriously harm a patient, the model catches nine in ten.

Eighty-six-point-three per cent accuracy, against a **seventy-six-point-one per cent** majority-class baseline. I put the baseline on the slide deliberately, because with classes this imbalanced, quoting eighty-six per cent alone would be misleading. The real improvement is ten points, and McNemar's test confirms it is not chance.

Macro AUC-ROC of point nine four three across the three tiers, which says the model's ranking is strong even where its thresholded decisions are not.

And then the two cards, which are the honest part.

It over-flags, on purpose. Precision on the High tier is **point six five eight** — so about a third of High alerts are not truly High. That is a deliberate trade. I tuned the decision threshold to favour recall, because in a decision-support tool a false alarm costs attention and a missed interaction can cost a patient.

The Low tier is weak, and rare. Just **four hundred and thirty-three of two hundred and eighteen thousand** pairs. The model handles it poorly. But a Low prediction raises no alert at all, so the practical cost of getting it wrong is close to nothing."

WHERE THE NUMBERS COME FROM

Every figure from `ml_pipeline/models/test_metrics.json`:

91.0% — `per_tier.high.recall` = 0.9104.

86.3% — `accuracy` = 0.8625.

76.1% — `baseline_majority_accuracy` = 0.7609.

0.943 — `auc_roc_macro_ovr` = 0.9433.

218,518 — `test_rows`.

0.658 — `per_tier.high.precision` = 0.6583.

433 — `per_tier.low.support` .

Additional figures you may want but which are not on the slide: macro-F1 is 0.5938; McNemar's chi-squared is 6,823.8; the decision thresholds are High ≥ 0.40 and Low ≥ 0.75 , tuned on the validation set; High-tier calibration improves from a Brier score of 0.0783 raw to 0.0677 after isotonic regression, with expected calibration error falling from 0.0661 to 0.0018.

IF THEY ASK

"Your macro-F1 is 0.594, which is not high. Why lead with accuracy and recall?" — Because macro-F1 is dragged down almost entirely by the Low tier, whose F1 is 0.113 on 433 examples. Moderate is 0.904 and High is 0.764. Macro-averaging treats those 433 pairs as one-third of the score. I report macro-F1 in the dissertation and on slide 24 precisely because it is the least flattering number, but the tier that carries clinical consequence is High, and that is what the headline reports.

"A third of your High alerts are false. Won't clinicians stop trusting it?" — That is the alert-fatigue risk and it is real; the literature in my Chapter Two finds override rates around ninety per cent for interaction alerts generally. My defence is that the threshold is a tunable parameter, not a fixed property — it is set at 0.40 and could be raised to trade recall for precision once there is evidence about what practitioners tolerate. Choosing that operating point properly needs the pharmacist review in recommendation four.

"How is a probability threshold of 0.40 justified?" — It was tuned on the validation set to maximise macro-F1, not chosen by hand, and the test set was untouched during that tuning. The asymmetry — High at 0.40, Low at 0.75 — encodes the safety asymmetry directly: it is easy to be classified High and hard to be classified Low.

"Is the model well calibrated?" — Not out of the box. Raw expected calibration error on the High tier is 0.066. After isotonic regression fitted on the validation set it drops to 0.002, and the Brier score improves from 0.078 to 0.068. So the ranking was always good; the probabilities needed correcting before they could be read as probabilities.

"218,518 test pairs sounds large. Are they independent?" — Not fully, and this is a genuine caveat. Pairs sharing a drug share features, so the effective sample size is smaller than the raw count. That is precisely why the drug-disjoint evaluation on slide 24 exists — it is the split that removes that dependence, and performance drops when it does.

Slide 23 — Results: feature importance and errors

WHAT IS ON THE SLIDE

Two charts side by side, with a dark summary strip beneath.

Left — Top 6 feature importances, as a vertical bar chart. Risk score product **0.197**; risk score minimum **0.182**; risk score maximum **0.116**; Drug A risk score **0.104**; Drug B risk score **0.095**; ATC class missing **0.082**. The first five bars are in full teal, the sixth in a lighter tone. Caption: Gini importance · these 6 of 27 carry 77.5%.

Right — Confusion matrix, a three-by-three heat map with true tiers as rows and predicted tiers as columns, cells shaded by row proportion:

True Low (n = 433): 63 (14.5%) · 299 (69.1%) · 71 (16.4%)

True Moderate (n = 166,273): 616 (0.4%) · 141,249 (85.0%) · 24,408 (14.7%)

True High (n = 51,812): 0 (0.0%) · 4,642 (9.0%) · 47,170 (91.0%)

Caption: row-normalised · 218,518 held-out drug pairs.

The strip: errors run upward. 24,408 Moderate pairs were graded High, while under 0.2% of High pairs fell to Low — the asymmetry that buys 91.0% High-tier recall.

WHAT TO SAY

Ninety seconds. Two charts; take them in order and do not blur them together.

"This slide answers two questions: what does the model lean on, and how does it get things wrong.

On the left, **feature importance**. And the striking result is that the **top five features are all the same thing** — they are all measures of each drug's recorded risk history. The product of the two risk scores, the minimum, the maximum, and each drug's individual score. Together those five carry **sixty-nine per cent** of the model's total importance. Add the sixth and these six of twenty-seven carry **seventy-seven-point-five per cent**.

That has two consequences, and they point in opposite directions. It makes the model **transparent** — I can tell you exactly why it graded a pair the way it did. And it makes it **fragile on new drugs**, which is the next slide.

The sixth bar is worth a moment: **a missing ATC class is itself predictive**. The model has partly learned which records are incomplete, not only which drugs interact. That is a data-quality signal leaking into the model and I would not defend it as a feature I wanted.

On the right, the **confusion matrix**, normalised by row. Read the diagonal: **fourteen-point-five per cent** on Low, **eighty-five per cent** on Moderate, **ninety-one per cent** on High.

And read the shape of the errors, which is what the strip says. **Twenty-four thousand Moderate pairs were pushed up to High**. Almost nothing was pushed down. The errors run **upward**, toward caution — and that asymmetry is not an accident, it is what the threshold tuning bought, and it is what produces the ninety-one per cent recall on the tier that matters."

WHERE THE NUMBERS COME FROM

The bar chart — `ml_pipeline/models/feature_importance.csv`, the top six of twenty-seven rows, verbatim. The importances are Gini (mean decrease in impurity), which is what scikit-learn's `feature_importances_` returns for a Random Forest. Top five sum to 0.6937; top six to 0.7753.

The confusion matrix — reconstructed from the per-tier precision, recall and support in `test_metrics.json`. Those three quantities fix every row total, every column total and the entire diagonal, and the diagonal divided by the total gives 0.8625, which matches the reported accuracy exactly. One degree of freedom remains, bounded to 71 pairs out of 218,518, which is why the True High → Predicted Low cell is shown as 0 and why the strip says "under 0.2%" rather than "none".

Be ready to say this plainly. The training script does not currently write a confusion matrix; adding one is a two-line change that would make this first-party for the final submission.

IF THEY ASK

"Gini importance is known to be biased toward high-cardinality features. Does that undermine this?" —

It is a fair objection. Gini importance can inflate continuous, high-cardinality features relative to binary ones, and the risk-score features are continuous while many of the others are flags. A permutation importance analysis would be the proper check and I did not run one. What supports the conclusion independently is the cold-start result on the next slide: if the risk-history features were not genuinely carrying the model, removing the drug's history would not have cost fourteen points of macro-F1.

"Isn't 'recorded risk history' circular? You are predicting interaction severity from interaction history."

— This is the sharpest question on the slide and it deserves a direct answer. It is not circular for the specific pair being scored — the risk score for drug A is computed from A's interactions with *other* drugs, never from the pair under evaluation, so the label is not leaking into its own features. But it is fair to say the model is learning "drugs that are dangerous in general are dangerous together", which is a weaker claim than "this specific mechanism produces this specific interaction". The cold-start evaluation is the measurement of exactly how much of the performance rests on that weaker claim.

"Why is the Low row so bad?" — Four hundred and thirty-three examples out of two hundred and eighteen thousand. The model sees roughly one Low pair for every four hundred Moderate ones, even with balanced class weighting. Most true Low pairs are predicted Moderate, which is the conservative direction. The literature reports the same problem: severity datasets are dominated by moderate cases, and my Chapter Two cites it as a known unsolved difficulty rather than an artefact of my data.

"Fourteen-point-seven per cent of Moderate pairs graded High. That is twenty-four thousand false alarms." — It is, and it is the cost side of the trade the slide describes. In exchange, only nine per cent of genuinely High pairs are missed. If I had tuned for balanced error the over-flagging would fall and the missed-High rate would rise, and I judged that the worse outcome for a safety tool. It is a defensible choice rather than an inevitable one, and I would say so.

Slide 24 — Results: cold start evaluation

WHAT IS ON THE SLIDE

Two panels side by side.

CATALOGUED DRUGS — accuracy 86.3%, macro F1 0.594, High-tier recall 0.910.

DRUGS HELD OUT OF TRAINING ENTIRELY — accuracy 69.6%, macro F1 0.472, High-tier recall 0.864.

A full-width card beneath: the five strongest features are all measures of a drug's recorded risk history. That makes the model transparent and auditable, and it is exactly why a genuinely new medicine, with no history to draw on, is predicted less reliably.

WHAT TO SAY

Seventy seconds. Volunteer this rather than being drawn to it.

"This is the slide I would put to myself if I were on the panel, so I want to present it rather than have it extracted.

On the left, the catalogued case — the numbers from two slides ago. On the right, the **drug-disjoint** split: every drug in the test set has been held out of training entirely, so every test pair involves a medicine the model has genuinely never seen. It simulates a newly approved drug.

Accuracy falls from **eighty-six-point-three to sixty-nine-point-six per cent**. Macro-F1 from **point five nine four to point four seven two**. That is a real and substantial degradation and I am not going to soften it.

But look at the third row. **High-tier recall falls only from point nine one zero to point eight six four**. So the model loses a great deal of its overall precision on unfamiliar drugs, while retaining most of its ability to catch the dangerous cases. For a safety tool, that is the more forgiving of the two failure modes.

And the card at the bottom is the causal explanation, which connects directly to the previous slide. The top five features are all a drug's recorded risk history. That is precisely what makes the model transparent — and precisely why a drug with no history is predicted less reliably. **The transparency and the fragility are the same property**. I would not claim one without the other."

WHERE THE NUMBERS COME FROM

Left panel from `test_metrics.json` (regime: random). Right panel from `cold_start_metrics.json` (regime: coldstart): accuracy 0.6958, f1_macro 0.4719, high_recall 0.8641.

The cold-start split used 941,950 training rows and 410,160 test rows. The same three algorithms were compared on it and the Random Forest won again (0.5718 macro-F1 on validation, against 0.5591 and 0.4838). McNemar's chi-squared is 5,038.7. The cold-start majority baseline is 77.2%, so at 69.6% accuracy the model is **below** the baseline on this split — a point worth knowing before someone else notices it.

IF THEY ASK

"**Your cold-start accuracy of 69.6% is below the 77.2% majority baseline. Doesn't that mean the model is worse than guessing?**" — On raw accuracy, yes, and I should say so plainly. But accuracy is the wrong measure here: the baseline achieves 77.2% by predicting Moderate for everything, which means it catches **zero** High-risk interactions. My model catches 86.4% of them. A model that finds nine in ten dangerous pairs at the

cost of over-flagging is more useful than one that is more often right and never warns anyone. This is the clearest case in the whole project for why accuracy alone is a misleading metric.

"How would you fix the cold-start weakness?" — Two routes. Add features that do not depend on history — molecular structure descriptors or target-binding profiles, which are available for a new drug on the day it is approved. Or use a pre-trained chemical representation. Both add computational weight, which conflicts with the on-device constraint, so it is a genuine trade rather than an oversight.

"Is the drug-disjoint split too harsh? Real deployments see a mix." — It is deliberately the hardest case: it assumes every drug is new. Reality sits between the two panels, so the honest reading is that these are bounds rather than a point estimate. I present both because reporting only the random split would overstate what the model does in practice.

"Did you also check calibration on the cold-start split?" — Yes, and it degrades too. Expected calibration error on the High tier goes from 0.002 on the random split to 0.038 on cold start after the same isotonic correction. So the probabilities are less trustworthy on unfamiliar drugs as well as the decisions.

Slide 25 — Results: deployment and testing

WHAT IS ON THE SLIDE

Three statistic cards:

7,984 bytes — the whole distilled on-device model.

95.6% of its decisions match the full Random Forest.

34 automated tests, including ten named clinical drug pairs.

Two cards beneath. The first: warfarin with ibuprofen, simvastatin with clarithromycin, sildenafil with nitroglycerin — real pairs with known clinical outcomes. The second: a dedicated offline group confirms the rule-based checks still work with no network, which verifies the study's central architectural claim.

WHAT TO SAY

Sixty-five seconds.

"Deployment, and whether any of it actually works.

Seven thousand nine hundred and eighty-four bytes. That is the entire on-device model — the distilled network that runs when there is no connection. Eight kilobytes.

It agrees with the full Random Forest on **ninety-five-point-six per cent** of decisions. Distillation costs something, and that is the measured cost: four and a half per cent disagreement, and High-tier recall of point eight eight against the forest's point nine one.

Thirty-four automated tests, and among them **ten named clinical drug pairs** with known outcomes — warfarin with ibuprofen, simvastatin with clarithromycin, sildenafil with nitroglycerin. Those are not synthetic cases; they are combinations with documented clinical consequences, and the application is asserted to grade each one correctly.

And the last card is the one that matters most to the thesis. There is a **dedicated offline test group** that runs the safety checks with the network disabled. That group is what turns 'works offline' from a claim into a tested property. It is the central architectural argument of this study, and I did not want it resting on assertion."

WHERE THE NUMBERS COME FROM

7,984 bytes and **95.6%** — `ml_pipeline/models/parity_report.json`: `tflite_bytes: 7984`, `agreement_rf_mlp_with_rule: 0.9564`. The same file records `mlp_high_recall_with_rule: 0.8803`, the distilled model's High-tier recall.

Distillation method — soft-label: the student network was trained on the forest's `predict_proba` output rather than on hard class labels. Note that an earlier draft of section 3.8.9 showed `rf.predict` in its code excerpt, which contradicted the surrounding text; that was corrected to `predict_proba` in the August revision.

34 tests — twelve test functions in `backend/tests/test_api.py`, plus twenty-two mobile test files. Those twenty-two files contain 165 individual test cases, so the true count of assertions is far higher; the slide figure is conservative. Know this breakdown before the day.

IF THEY ASK

"**Thirty-four tests is not many for a system this size.**" — The count is files plus functions, not individual cases. The twenty-two mobile files contain a hundred and sixty-five separate test cases, and the backend twelve

are twelve functions. So the real figure is close to a hundred and eighty. The slide is understated rather than inflated.

"What do the backend tests actually cover?" — Health and root routes, the database version and delta endpoints, prediction-endpoint validation and success shape, unknown-drug handling, rate limiting at thirty requests per minute, authentication enforcement, and audit-log hash-chain tamper detection.

"Ten drug pairs is a very small clinical validation." — It is not a clinical validation and I would not describe it as one. It is a regression test: a set of well-documented pairs whose expected grade is known, used to catch a change that breaks the safety logic. Clinical validation is recommendation four, and it needs licensed pharmacists.

"Ninety-five-point-six per cent agreement — what happens in the other four-point-four?" — Mostly boundary cases near the decision thresholds, where the student network and the forest fall on opposite sides. The consequential number is High-tier recall: 0.880 for the distilled model against 0.910 for the forest. So the offline path is measurably more likely to miss a High interaction, which is exactly why the cloud model is preferred whenever a network exists.

"Did you measure inference latency?" — Yes, per-request on the API and with on-device timing tests on the client, against the targets set in Chapter Three. But single-request only. Concurrent load is untested, and that is limitation three on slide eight.

Slide 26 — Results: usability testing

WHAT IS ON THE SLIDE

A large left card carrying the headline: **84.3** mean System Usability Scale score, 16.3 points above the 68-point benchmark for an average system. Beneath a rule: 74 of 75 task scenarios completed · 71 with no assistance at all.

A right card holds a small bar chart of the three participant groups:

Physiotherapy students (5) — **96.0**

Nursing students (6) — **89.2**

Clinic staff (4) — **62.5**

And a card beneath it: the split is not about drug knowledge. Students tested in a quiet room; clinic staff tested during a working day. Every instance of a participant needing help came from that group, and all of it on one task — finding the allergy profile.

WHAT TO SAY

Eighty seconds. The second half of this slide is more valuable than the first; do not spend all your time on the headline.

"Fifteen participants, five task scenarios each, and the standard ten-item System Usability Scale.

The mean is **eighty-four-point-three**, which is **sixteen points above** the sixty-eight-point benchmark for an average system. **Seventy-four of seventy-five** task scenarios were completed, **seventy-one of them with no assistance at all**.

But the aggregate hides the finding, and the finding is on the right.

Ninety-six for physiotherapy students. **Eighty-nine** for nursing students. **Sixty-two-point-five** for clinic staff. That is a **thirty-three point spread**, which is far too large to ignore.

My interpretation — and I want to be clear it is an interpretation, not a measured cause — is that **the split is not about drug knowledge**. The students tested in a quiet room with nothing else to do. The clinic staff tested during a working day, with interruptions.

And two facts support that. **Every single instance of a participant needing help came from that group**. And **all of it was on one task**: finding the allergy profile.

So the finding is not 'the app is hard for clinic staff'. It is 'one screen is hard to find when you are busy'. That is a specific, actionable defect, and it is recommendation one on the next slide."

WHERE THE NUMBERS COME FROM

Mean SUS **84.33**, standard deviation **15.82**, range 50.00 to 100.00. Group means: physiotherapy students 96.00 (SD 7.62, range 82.50–100.00); nursing students 89.17 (SD 7.19, range 75.00–95.00); clinic staff 62.50 (SD 9.79, range 50.00–72.50). Table 4.9 of the final dissertation.

Weighted check: $(5 \times 96.00 + 6 \times 89.17 + 4 \times 62.50) \div 15 = 84.33$. The subgroup means reconcile to the overall mean exactly.

The 68-point benchmark and the interpretation bands come from Bangor, Kortum and Miller (2008). Note that on their scale 84.33 sits in the **"Good"** band (72–84), not "Excellent" (≥ 85). An earlier version of the usability

report described it as Excellent; that was corrected. If you say anything, say "Good, at the top of the band" — it is accurate and it is more credible than an overclaim.

Task completion: 71 of 75 unassisted (94.7%), 74 of 75 achieved including assisted (98.7%). The four non-clean outcomes were three assisted completions and one failure, all on the allergy-profile task.

IF THEY ASK

"Fifteen participants is a small sample." — It is, and the confidence interval around 84.3 is wide. It exceeds the five-to-ten specified in my Chapter Three and sits above the commonly cited minimum of twelve for a stable SUS mean, but I would not present it as a precise population estimate. The subgroup analysis, with four and five people per group, is weaker still and I present it as a signal to investigate rather than as an established effect.

"You claim environment rather than skill caused the gap, but you did not manipulate environment. How do you know?" — I do not know; it is an inference and I would label it as such. What supports it is that all assisted completions came from that one group and all of them were on one task, which is a pattern more consistent with attention than with capability. What would settle it is testing students under interruption, and I did not do that.

"Why no pharmacists?" — Recruitment access, honestly. It is limitation two on slide eight, and it is the most important gap in the evaluation because practitioners are half the intended audience.

"Is 84.3 Good or Excellent?" — Good, at the top of the band. Bangor's Excellent threshold is 85.

"One participant scored 50, which is below average. What happened there?" — That participant was in the clinic-staff group and was the one who did not complete the allergy task. It is the same defect showing up in both measures, which is part of why I am confident the allergy screen is the specific problem rather than the application generally.

Slide 27 — Conclusion and recommendations

WHAT IS ON THE SLIDE

A full-width conclusion statement: an offline-first medication safety tool, driven by a machine-learning severity model, can be built from freely licensed data and software — and used by the people it is meant for. MedGuard is a functioning academic prototype, not a clinically validated product.

Four recommendation cards beneath:

- 1 — Prompt for allergies during onboarding, the one fix the usability data asks for.
- 2 — Run a scripted load test against the API and the on-device model.
- 3 — Rebuild the reference database as soon as DrugBank restores downloads.
- 4 — Structured review by licensed pharmacists before any wider use.

WHAT TO SAY

Sixty seconds. This is your closing argument; slow down and land the last sentence.

"The conclusion, in one sentence: an offline-first medication safety tool, driven by a machine-learning severity model, **can** be built from freely licensed data and software — and used by the people it is meant for.

And the second sentence matters as much as the first. **MedGuard is a functioning academic prototype, not a clinically validated product.** I would rather say that myself than have it said to me.

Four recommendations, and each one comes from a specific finding rather than from general good practice.

One. Prompt for allergies during onboarding. That is the single fix the usability data asks for — one task, one group, every assisted completion.

Two. Run a scripted load test against the API and the on-device model. Limitation three; I know the latency of one request and nothing about concurrency.

Three. Rebuild the reference database as soon as DrugBank restores academic downloads. That closes the fourteen per cent of records that lag the model.

Four. Structured review by licensed pharmacists before any wider use. That is the precondition for everything else — for a clinical claim, for more directive wording, and for choosing the alert threshold on evidence rather than on my judgement."

WHERE THE NUMBERS COME FROM

Each recommendation maps to a limitation on slide 8 or a finding on slide 26. Recommendation one comes from the allergy-task failures; two from the untested concurrency; three from the 13.9% stale records; four from the absence of any clinical or regulatory review.

IF THEY ASK

"**What is the single most important next step?**" — Recommendation four, pharmacist review. Everything else is engineering that I know how to do. Pharmacist review is the thing that changes what the system is allowed to claim, and without it the tool cannot move beyond prototype no matter how good the code is.

"Would you deploy this today if you could?" — No. Not because it does not work, but because a third of High alerts are false positives and no clinician has assessed whether that is tolerable. Deploying a safety tool without that assessment would be the opposite of safe.

"What would you do differently if you started again?" — Two things. I would build the cold-start evaluation first rather than last, because knowing that history features dominate would have changed the feature design from the beginning. And I would recruit practitioners before students, because the practitioner cohort is the one the evaluation is missing and it was the harder one to arrange.

Slide 28 — References

WHAT IS ON THE SLIDE

Twelve sources in two columns. The full reference list in the dissertation runs to thirty-six entries; these are the ones cited in the presentation itself.

Bangor, Kortum & Miller (2008) — SUS empirical evaluation and interpretation bands.

Breiman (2001) — Random forests, the original paper.

Brooke (1996) — the SUS instrument.

Davis (1989) — the Technology Acceptance Model.

Delara et al. (2024) — polypharmacy prevalence umbrella review.

Ekpenyong et al. (2018) — Nigerian pharmacy workforce analysis.

Hevner et al. (2004) — Design Science Research.

Knox et al. (2024) — DrugBank 6.0.

Oriaifo et al. (2024) — interaction predictors at UBTH, Benin City.

Sutton et al. (2020) — overview of clinical decision support systems.

Wang et al. (2024) — ML for DDI prediction, comprehensive review.

World Health Organisation (2021) — Global patient safety action plan 2021–2030.

WHAT TO SAY

Ten seconds. Do not read them.

"Sources for everything cited. The full reference list in the dissertation runs to thirty-six entries; these are the twelve cited in this presentation."

WHERE THE NUMBERS COME FROM

The dissertation's reference list was reconciled in the August revision: six citations that appeared in the text without a reference-list entry were resolved, every DOI was verified against Crossref, and the list now stands at thirty-six entries with every one cited and every citation resolved.

IF THEY ASK

"Why is Davis (1989) here? The Technology Acceptance Model does not appear in your presentation." — It underpins the usability evaluation. The System Usability Scale measures the perceived-ease-of-use construct from TAM; perceived usefulness is not captured by the scale and was approached indirectly through task completion and the post-session debrief. It is cited in Chapter Three rather than on a slide.

"Sutton et al. (2020) — where does that appear?" — Chapter Two, for the general framing of clinical decision support systems and the evidence on alert fatigue and override rates.

"Breiman (2001) for Random Forests — did you use anything beyond the original formulation?" — No. Balanced class weighting and post-hoc threshold tuning, both standard, and isotonic calibration from a separate literature. The forest itself is Breiman's, as implemented in scikit-learn.

Slide 29 — Thank you

WHAT IS ON THE SLIDE

A closing slide mirroring the title slide's layout. The MedGuard mark and wordmark at the top left, then **Thank you**, then "Questions welcome." Below a rule: your name and matriculation number, your supervisor, and the department and institution. The right third carries the application screenshot panel.

WHAT TO SAY

Five seconds. Then stop talking and let them ask.

"Thank you. I am happy to take questions."

WHERE THE NUMBERS COME FROM

Not applicable.

IF THEY ASK

This is where the general questions land. Four that are not tied to any single slide:

"What is your original contribution?" — Not the algorithm; Random Forests are from 2001. The contribution is the demonstrated pipeline: severity labels derived from pharmacological features rather than editorial consensus, a model carried all the way from training through an API to an eight-kilobyte on-device artefact, an entire reference database shipped inside a mobile application, and the whole thing evaluated with real users. Wang et al. identify that last step as the one the field consistently does not take.

"If you had three more months, what would you do?" — Recruit fifteen practising pharmacists and repeat the usability study, because that is the cohort the evaluation is missing. In parallel, add structure-based features to attack the cold-start weakness, and dump a proper confusion matrix and a permutation-importance analysis from the training script.

"Is machine learning necessary here? Could a rule engine have done it?" — For catalogued pairs, a rule engine is what I use: the curated grade is returned directly, no model involved. The model exists for pairs that no rule covers, which is where a rule engine returns nothing at all. So the answer is that both are used, and the model handles the case the rules cannot reach.

"How do you know the model is not simply memorising DDInter?" — Because the drug-disjoint split removes that possibility for the drugs concerned, and performance drops but does not collapse — High-tier recall stays at 0.864. If it were pure memorisation, that number would fall to chance.

Appendix A — Master number sheet

Every figure that appears in the deck, with its source. If a number is challenged, this is where you look.

FROM ML_PIPELINE/MODELS/TEST_METRICS.JSON (RANDOM SPLIT)

FIGURE	VALUE	FIELD
Accuracy	0.8625 → 86.3%	accuracy
Majority baseline	0.7609 → 76.1%	baseline_majority_accuracy
Macro F1	0.5938 → 0.594	f1_macro
Macro AUC-ROC	0.9433 → 0.943	auc_roc_macro_ovr
Test rows	218,518	test_rows
Training rows	1,019,738	train_rows
High recall	0.9104 → 91.0%	per_tier.high.recall
High precision	0.6583 → 0.658	per_tier.high.precision
High F1	0.7641	per_tier.high.f1
High support	51,812	per_tier.high.support
Moderate precision / recall / F1	0.9662 / 0.8495 / 0.9041	per_tier.moderate
Moderate support	166,273	per_tier.moderate.support
Low precision / recall / F1	0.0928 / 0.1455 / 0.1133	per_tier.low
Low support	433	per_tier.low.support
Decision thresholds	High ≥ 0.40 , Low ≥ 0.75	decision_thresholds
McNemar chi-squared	6,823.8	mcnemar_chi2
Brier, High tier	0.0783 raw → 0.0677 isotonic	calibration_high_tier
ECE, High tier	0.0661 raw → 0.0018 isotonic	calibration_high_tier
Model comparison (validation macro-F1)	RF 0.5615 · HGB 0.5484 · LR 0.4858	model_comparison_val_macro_f1

FROM ML_PIPELINE/MODELS/COLD_START_METRICS.JSON (DRUG-DISJOINT SPLIT)

FIGURE	VALUE
Accuracy	0.6958 → 69.6%
Macro F1	0.4719 → 0.472
Macro AUC-ROC	0.8507
High recall	0.8641 → 0.864
High precision	0.4195
Majority baseline	0.7716 → 77.2%
Training / test rows	941,950 / 410,160
McNemar chi-squared	5,038.7
ECE, High tier	0.1461 raw → 0.0381 isotonic
Model comparison	RF 0.5718 · HGB 0.5591 · LR 0.4838

FROM ML_PIPELINE/MODELS/PARITY_REPORT.JSON

FIGURE	VALUE
On-device model size	7,984 bytes
Agreement with Random Forest	0.9564 → 95.6%
Distilled model High recall	0.8803
Distillation method	soft-label, trained on predict_proba

FROM ML_PIPELINE/MODELS/FEATURE_IMPORTANCE.CSV (27 FEATURES, SUMS TO 1.0)

RANK	FEATURE	IMPORTANCE
1	drug_risk_prod	0.19671
2	drug_risk_min	0.18216
3	drug_risk_max	0.11573
4	drug_risk_a	0.10369
5	drug_risk_b	0.09543
6	atc_missing_either	0.08156
7	half_life_ratio	0.04841
8	half_life_missing	0.03651
9	drug_type_match	0.03274
10	category_overlap	0.02955
—	Top 5 combined	0.69372 → 69.4%
—	Top 6 combined	0.77528 → 77.5%
—	Remaining 22	0.30628 → 30.6%

FROM THE SHIPPED DATABASE MOBILE/ASSETS/DB/MEDGUARD.DB

TABLE	ROWS
drugs	4,955
interactions_data	851,868
drug_synonyms	252,046
drug_categories	65,784
drug_enzymes	20,822
food_interactions	2,503
tmpl_lookup / mech_lookup / effect_lookup	718 / 15 / 13
File size	75.1 MiB = 78.75 MB → reported as 79 MB

FROM THE USABILITY STUDY (TABLE 4.9)

GROUP	N	MEAN SUS	SD	RANGE
Physiotherapy students	5	96.00	7.62	82.50–100.00
Nursing students	6	89.17	7.19	75.00–95.00
Clinic staff	4	62.50	9.79	50.00–72.50
All participants	15	84.33	15.82	50.00–100.00

Task completion: 71 of 75 unassisted (94.7%); 74 of 75 achieved (98.7%). Benchmark for an average system: 68 (Bangor et al., 2008); "Good" band 72–84, "Excellent" ≥ 85.

FROM THE LITERATURE

FIGURE	SOURCE
1 in 30 harmed; \$42bn; 50% of preventable harm	WHO (2021, 2023)
Polypharmacy 37% general, 45% older adults	Delara et al. (2024)
1,413,413 interactions, ~11,900 drugs	Knox et al. (2024), DrugBank 6.0
0.53–0.66 pharmacists per 10,000	Ekpenyong et al. (2018)
50.5% on admission, 67.4% at discharge, n = 430	Oriaifo et al. (2024)
Medscape 202 / 400 accuracy score	comparative study cited in Chapter Two
DDInter ~240,000 interactions	Xiong et al. (2022); Tian et al. (2025)

Appendix B — The six questions most likely to be asked

Ranked by likelihood, with the shortest defensible answer.

1. "Your cold-start accuracy is below the majority baseline. Is the model any use?"

On accuracy, the baseline wins — by predicting Moderate for everything and catching zero High-risk interactions. My model catches 86.4% of them. Accuracy is the wrong measure for an imbalanced safety problem, which is why I report per-tier recall alongside it.

2. "A third of your High alerts are false positives. Won't that cause alert fatigue?"

It is a deliberate trade and the threshold is tunable. A false alarm costs attention; a missed High interaction can cost a patient. Choosing the right operating point needs practitioner evidence, which is recommendation four.

3. "Aren't your top features circular — predicting interaction risk from interaction history?"

Not for the pair being scored: a drug's risk score is computed from its interactions with other drugs, never the pair under evaluation. But the model is learning a general-danger signal rather than a mechanism-specific one, and the cold-start slide is the measurement of how much rests on that.

4. "Fifteen participants, none of them pharmacists. How much does the usability result tell us?"

It tells us the interface works for health-adjacent non-experts and that one screen is hard to find under interruption. It tells us nothing about practitioners, which is stated as a limitation rather than glossed.

5. "Why not use the best-performing method in the literature?"

Because graph neural networks cannot run on the target device. Deployability was a design constraint from the start, not an afterthought, and the research question is about what can be deployed offline rather than what scores highest on a benchmark.

6. "What is genuinely new here?"

Not the algorithm. The pipeline: severity derived from pharmacology rather than editorial consensus, carried from training through an API to an eight-kilobyte on-device model, with the entire reference database shipped inside the app, evaluated with real users. That last step is the one the literature reports is usually missing.